

| PROJECT PROPOSAL – AUGUST 20, 2026 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| NAME                               | ERA DUMANGCAS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| PROJECT TITLE<br>(specific)        | Arpie: A Context-Aware Endpoint Network Intrusion Detection and Threat-Response System for Public Wi-Fi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| SUBJECT:<br>CODE:<br>TIME:         | INFORMATION ASSURANCE AND SECURITY 2<br>5938<br>8:00-10:00AM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PROJECT NAME                       | #4 Network Intrusion Detection System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Technologies and Framework:        | <p><b>Programming Language</b></p> <ul style="list-style-type: none"> <li>- Python</li> </ul> <p><b>Network Analysis and Security Libraries</b></p> <ul style="list-style-type: none"> <li>- Scapy (Python)</li> <li>- psutil (Python)</li> </ul> <p><b>Application Framework</b></p> <ul style="list-style-type: none"> <li>- Flet (Python)</li> </ul> <p><b>Threat Intelligence</b></p> <ul style="list-style-type: none"> <li>- AbuseIPDB (API)</li> <li>- IPinfo Lite (API)</li> </ul> <p><b>Database</b></p> <ul style="list-style-type: none"> <li>- SQLite</li> </ul> <p><b>Testing</b></p> <ul style="list-style-type: none"> <li>- pytest (Python)</li> <li>- PCAP Files</li> </ul> |
| Detection Techniques               | <ol style="list-style-type: none"> <li><b>1. ARP Identity Inconsistency</b><br/>One local IP associated with more than one MAC address within a 5-minute window</li> <li><b>2. Port-Scan Behavior</b><br/>More than 15 unique destination ports contacted by one source within a 10-second window</li> <li><b>3. Traffic-Rate Anomaly</b><br/>SYN/UDP/ICMP packets from one source exceeding 100 packets/second (default baseline, configurable)</li> <li><b>4. Gateway Identity Change</b><br/>gateway MAC/IP association changes more than once within a 10-minute window</li> </ol>                                                                                                       |
| Development Tools                  | <ol style="list-style-type: none"> <li><b>1. Wireshark</b><br/>Analyze and verify captured packets</li> <li><b>2. Visual Studio Code</b><br/>Development environment</li> <li><b>3. Git/GitHub</b></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|                                      | <p>Version control and collaboration</p> <p><b>4. VirtualBox/VMware</b></p> <p>Create a safe testing environment</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Monitoring and Alert Features</b> | <ul style="list-style-type: none"> <li>• Network-context detection (trusted / public-untrusted / unknown) with adjustable alert thresholds</li> <li>• Real-time alerts with evidence, severity, confidence, and recommended action</li> <li>• IP reputation (AbuseIPDB) and geolocation/ASN (IPinfo Lite) enrichment for public IPs, cached locally and factored into the risk score</li> <li>• Transparent, heuristic session risk score (0–100)</li> <li>• Reversible "Seal Mode": user-confirmed, temporary firewall rule with Unseal/Restore and audit log</li> <li>• Event logging and exportable end-of-session report</li> </ul> |
| <b>Target Users</b>                  | <ul style="list-style-type: none"> <li>• Students who connect to public Wi-Fi on campus, in cafés, or in coworking spaces</li> <li>• Remote and hybrid workers who handle work traffic on public or unfamiliar networks</li> <li>• Small-Office/Home-Office (SOHO) users without a dedicated IT staff to monitor their network</li> </ul>                                                                                                                                                                                                                                                                                               |
| <b>User Roles</b>                    | <p><b>End User</b></p> <ul style="list-style-type: none"> <li>• View real-time dashboard and alerts</li> <li>• Confirm or dismiss Seal Mode actions</li> <li>• Export session reports</li> </ul> <p><b>Evaluator/Administrator</b></p> <ul style="list-style-type: none"> <li>• Load PCAP files and configure detection thresholds</li> <li>• Review event logs and session analytics</li> </ul>                                                                                                                                                                                                                                        |
| <b>LOGO/ICON</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Project Objectives:</b>           | <ol style="list-style-type: none"> <li>1. Classify the current network as trusted, public/untrusted, or unknown.</li> <li>2. Monitor local network traffic in real time using Scapy.</li> <li>3. Detects ARP spoofing, port scans, and abnormal traffic, enriched with AbuseIPDB and IPinfo Lite.</li> </ol>                                                                                                                                                                                                                                                                                                                            |

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|                             | <p>4. Respond through a reversible, user-confirmed "Seal Mode" and generate a session report.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Project Description:</b> | <p>Public Wi-Fi networks provide convenient connectivity for students, remote workers, and small-office/home-office users, but they also expose users to unfamiliar network environments where suspicious activity can be difficult to recognize. Arpie addresses this gap as a lightweight endpoint network intrusion detection application that combines network-context classification, passive packet monitoring, deterministic detection rules, threat-intelligence enrichment, and explainable alerts. Arpie, your Cyber-Detective Seal, helps users identify and respond to potential local-network threats through clear and understandable security information.</p> <p>The intended users are students, remote and hybrid workers, and small-office/home-office users who connect to public, shared, or unfamiliar networks without dedicated IT support. Arpie provides a clear, evidence-based assessment of network activity through alerts that include severity, confidence, supporting evidence, and recommended actions. This helps users make informed decisions about whether to remain connected or activate protective measures when suspicious activity is detected.</p> <p>The system is implemented in Python 3 using Scapy for packet capture and protocol inspection, psutil for process and network interface information, Flet for the desktop interface, and SQLite for event and session storage. AbuseIPDB provides public IP reputation information, while IPinfo Lite provides public IP geolocation and ASN information for network-context analysis. pytest and PCAP files support controlled and repeatable testing of the detection rules and system behavior.</p> <p>Arpie's MVP uses three deterministic detection rules: ARP identity inconsistency, port-scan behavior, and traffic-rate anomaly. Each detection reports supporting evidence and confidence. The system classifies networks as trusted, public-untrusted, or unknown and calculates a transparent 0 to 100 session risk score. Its reversible Seal Mode requires user confirmation before applying a temporary firewall rule, with automatic restoration after 30 minutes, manual Unseal/Restore, and an audit log. Event logging and an exportable end-of-session report provide a record of detected activity and response actions.</p> |
| <b>System Flow</b>          | <p>Detect Network Context → Select Monitoring Profile → Capture/Replay Traffic → Extract Evidence → Enrich with Threat Intelligence (AbuseIPDB/IPinfo Lite) → Apply Detection Rules → Score Severity &amp; Risk → Display Explainable Alert → Confirm Seal Mode → Log Event → Generate Session Report</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



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